





13. Propositional logic is also known as
 - A) First-order logic
 - B) Predicate logic
 - C) Boolean logic
 - D) Modal logic
14. A game tree is mainly used in
 - A) Constraint satisfaction
 - B) Two-player games
 - C) Machine learning
 - D) Bayesian networks
15. Which is NOT a component of propositional logic?
 - A) Connectives
 - B) Propositions
 - C) Quantifiers
 - D) Truth values
16. Resolution in logic is used for
 - A) Inference
 - B) Planning
 - C) Probability
 - D) Searching
17. Classical planning assumes that
 - A) The world is uncertain
 - B) The world is fully observable and deterministic
 - C) Actions are random
 - D) Multiple agents are always present
18. A drawback of sensor-less planning is
 - A) It cannot handle goals
 - B) The agent must plan for all possible situations
 - C) It cannot represent states
 - D) It eliminates search space
19. Backward search is typically used when
 - A) The initial state is not known
 - B) The goal state is partially observable
 - C) The goal state is explicitly defined
 - D) The state space is infinite
20. Hierarchical planning breaks down
 - A) Probabilities into components
 - B) Complex tasks into simpler sub-tasks
 - C) Search trees into nodes
 - D) Belief networks

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe the structure of an intelligent agent. Explain how the environment affects its performance.	8	L*2	1	1
	b) Compare Breadth First Search and Depth First Search algorithms with suitable illustrations.	8	L3	2	1
2.	a) Explain Hill-Climbing search and discuss its drawbacks.	8	L2	1	1
	b) Illustrate the structure of a goal-based agent with a diagram. How does it differ from simple reflex and utility-based agents?	8	L3	2	1
3.	a) Explain the role of heuristic functions in informed search. Illustrate with A* search.	8	L2	1	1
	b) Explain Iterative Deepening Depth First Search. Analyze how it combines the benefits of BFS and DFS while overcoming their limitations, with suitable examples.	8	L3	2	1
Unit – II					
4.	a) What is adversarial search? Explain the concept of a game tree with an example.	8	L2	3	1
	b) Discuss why handling uncertainty is important in AI reasoning. Analyze its impact on decision-making with suitable examples.	8	L3	4	1
5.	a) What is a Bayesian Belief Network? Explain its structure and working with a neat diagram.	8	L2	3	1
	b) Explain Alpha-Beta pruning and analyze how it improves the efficiency of the Minimax algorithm. Illustrate its working with a suitable game tree diagram.	8	L3	4	1

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| 6. | a) | Explain forward chaining and backward chaining with neat diagrams. Compare their approaches. | 8 | L2 | 3 | 1 |
| | b) | Analyze the working of the Minimax search algorithm. Illustrate with a suitable game tree and explain how the algorithm ensures optimal play. | 8 | L3 | 4 | 1 |

Unit – III

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| 7. | a) | What is classical planning in AI? Explain its key characteristics with examples. | 8 | L2 | 5 | 1 |
| | b) | Explain the concept of sensor-less planning in AI. Critically analyze its advantages and limitations. | 8 | L3 | 5 | 1 |
| 8. | a) | What is a planning graph? Describe its construction with an example. | 8 | L2 | 5 | 1 |
| | b) | Compare and contrast classical planning and hierarchical planning in AI. Analyze their differences in approach, efficiency, and applicability with suitable examples. | 8 | L3 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
